



# **Moon 101 Presentation**

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## **Moon 101**

### **Part 1 – Formation of the Moon**

# **Theories for Origin of Moon**

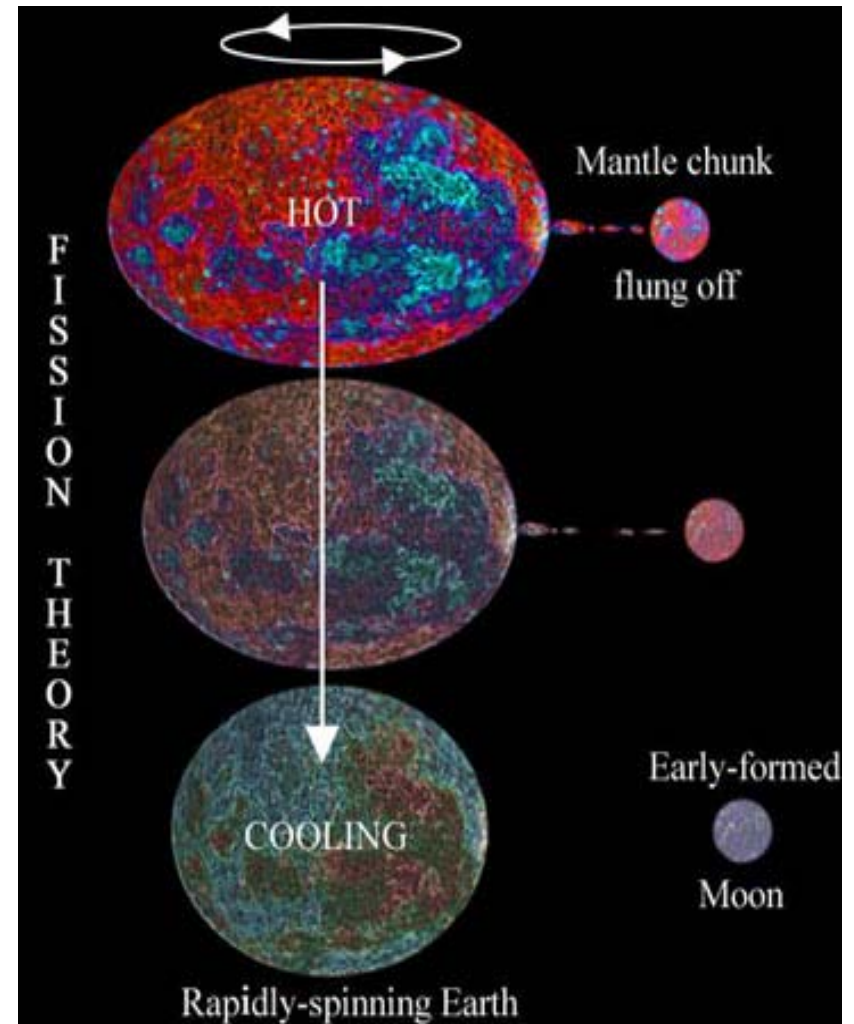
- Fission
- Co-creation
- Capture
- Impact

# Moon 101

## Part 1 – Formation of the Moon

Theories for Origin of Moon

**Fission Theory:**  
the Moon was  
originally part of  
the Earth but  
was torn free.





## **Moon 101**

### **Part 1 – Formation of the Moon**

#### **Theories for Origin of Moon** **Co-creation Theory:**

The two closely spaced proto-planets formed with the smaller one in orbit. They grew and eventually became what we know as Moon and Earth.





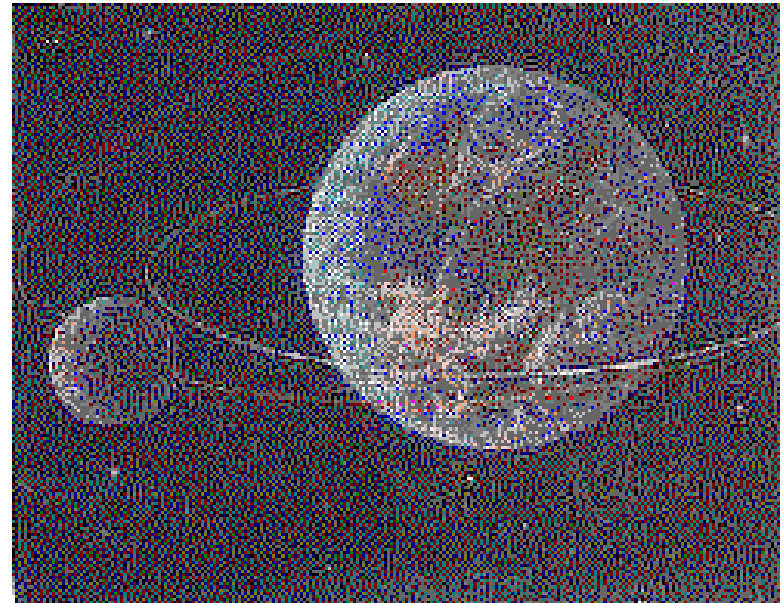
## **Moon 101**

### **Part 1 – Formation of the Moon**

Theories for Origin of Moon

## **Capture Theory:**

the Moon was formed somewhere else in the solar system and was later captured by the gravitational field of the Earth.



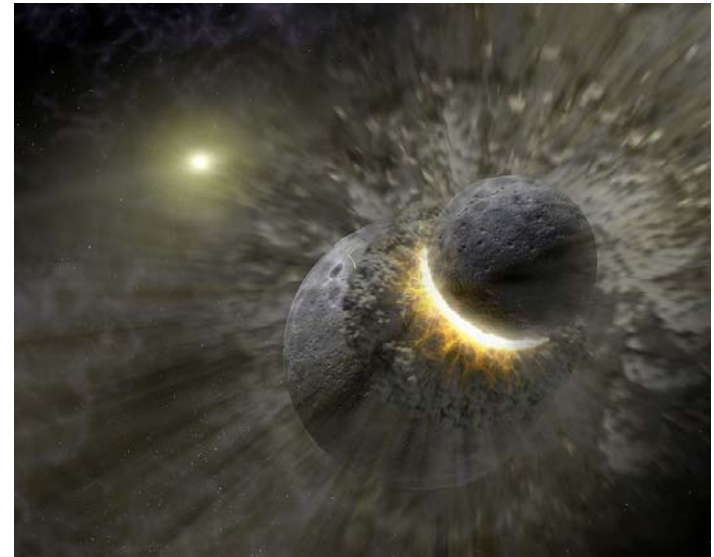


## **Moon 101**

### **Part 1 – Formation of the Moon**

#### **Theories for Origin of Moon**

**Impact Theory:** Earth was hit by a Mars' sized object. Most of the material reformed as the Earth, and some material orbited the Earth and eventually formed the Moon.



# **Moon 101**

## **Part 2 – Evolution of the Moon**

### **Features**

- Craters
- Rays
- Rilles
- Highlands
- Maria



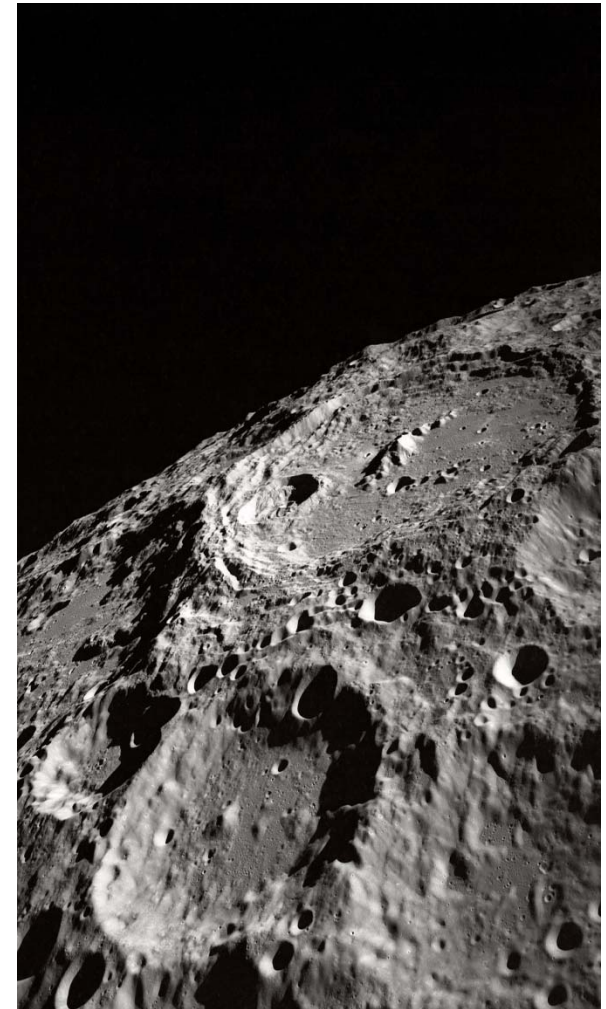


## Moon 101

### Part 2 – Evolution of the Moon

#### How They Formed

Primary and secondary **craters** were formed by meteor impacts which broke down and transported rock. This broken rock, called regolith or breccia, covers the surface of the Moon.





## Moon 101

### Part 2 – Evolution of the Moon

#### How They Formed

The **rays** surrounding the craters are material ejected during the impacts that formed the craters. Only the rays of the more “recently” formed craters maintain their features. Older rays have been degraded because of meteor impacts.

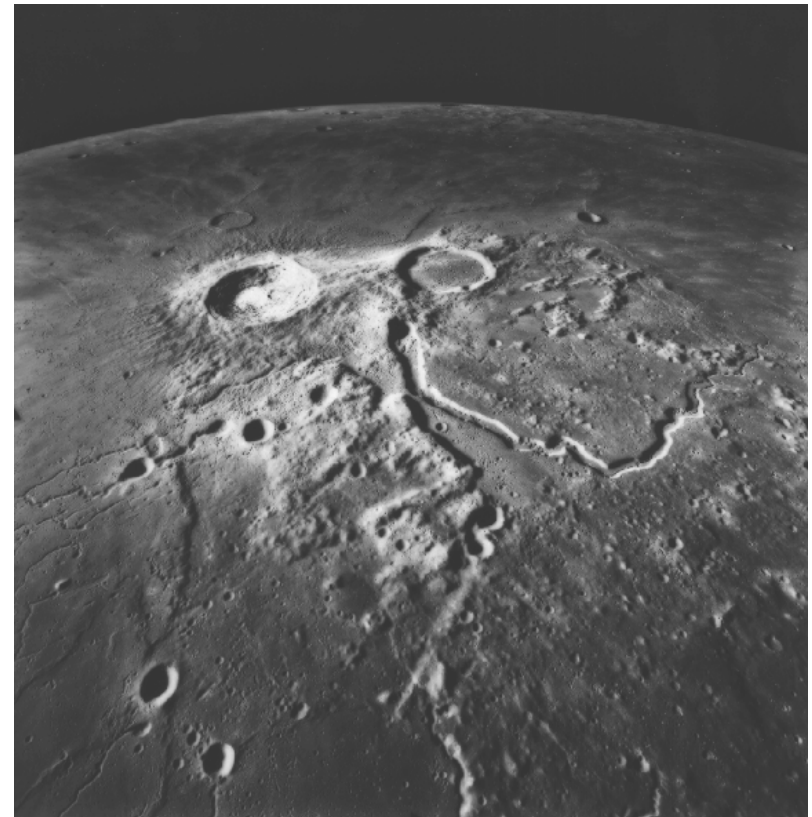


# Moon 101

## Part 2 – Evolution of the Moon

### How They Formed

**Rilles** are long, deep channels that run through the maria that are thought to be leftover lava channels from the formation of the maria.





## How They Formed

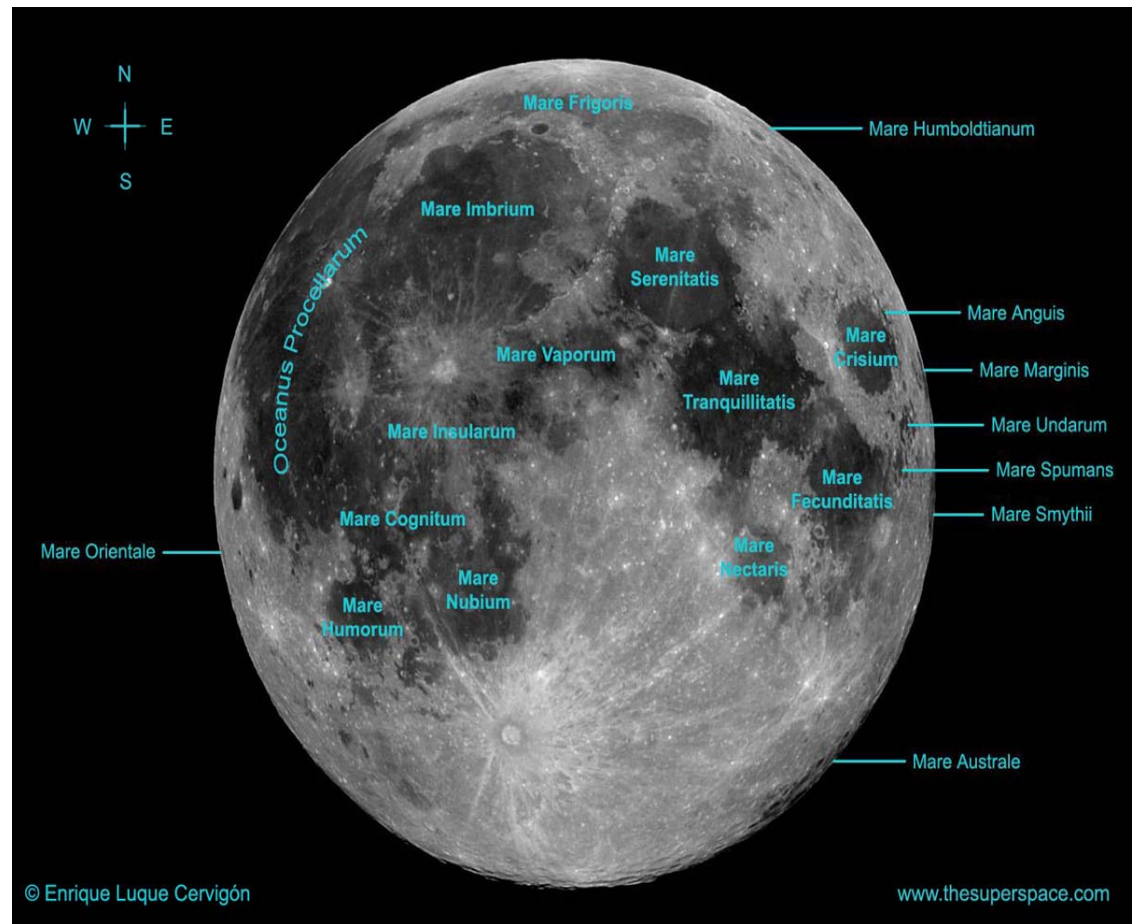
The **highlands** have the oldest surface rock and look like lunar mountain ranges. They were formed about 4.4 – 4.6 b.y. ago.



# Moon 101

## Part 2 — Evolution of the Moon

How They Formed  
Crater  
compressions  
filled with  
molten lava  
formed the  
**maria** basalt.





## **Moon 101**

### **Part 2 – Evolution of the Moon**

**Theories related to dating of “lunar cataclysm”**

**1. Lunar Cataclysm:** there were many bombardments in a short time period (3.9 B.Y. ago).

**2. Imbrium Impact:** all impact melt breccia were made by one gigantic event

**3. Stonewall Effect:** early bombardments degraded the impact melt breccias so the only fragments that have survived are from later impacts



## **Moon 101**

### **Part 2 – Evolution of the Moon**

#### **Testing the Lunar Cataclysm Theory**

- Using potassium argon dating, the oldest impact breccia appears to be 3.9 b.y. old
- In order to test the lunar cataclysm theory it would be reasonable to collect samples from maria significantly farther away from previous landing sites



## **Volcanoes on the Earth have more diversity than volcanoes on the Moon**

- Earth is more geologically diverse with plate tectonics
- Continents are actively moving
- There is a larger diversity of mineral types on Earth
- On Earth, surface rocks are drug down into subduction zones



## **Why are there more impact craters on the Moon than on the Earth?**

The Earth's atmosphere usually burns up the meteors before they reach the surface of the Earth.

The craters that have been created on Earth have been destroyed or obscured by weathering or sedimentation.